Individual Weekly Report – Week 7

Dimensionality Reduction

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Department	Artificial Intelligence and Data Science
Project Title	Youth Smoking and Drug Consumption: From Raw Data to Visual Insights
Role	Data Visualization and Dashboard Development
Week	Week 7 – Dimensionality Reduction

Objective of the Week

The aim of this week was to prepare the visual side of dimensionality reduction using PCA. PCA reduces a large number of variables into fewer components, and visualization helps to explain these results.

Work Done

Preparing PCA visualizations: I prepared visualizations of the PCA results so that the team could see how the reduced components represent the data.

Studying ways to present the results: I studied different ways of presenting dimensionality reduction results and compared which would be the easiest for a general audience to understand.

Planning the dashboard design for PCA outputs: I planned how the PCA outputs would be shown in the dashboard, including where they would be placed and how they would connect with the other charts.

Organizing important visualization results: I organized all the important visualization results collected so far. This made it easier to include them in the dashboard in the next week.

Tools and Techniques Used

PCA result visualizations, study of presentation methods and dashboard planning.

Outcome of the Week

The PCA results were prepared in a visual form, and a clear plan existed for showing them in the dashboard.

What I Learned

I learned that technical results like PCA are difficult to understand in numbers alone. Good visualization makes them much easier to follow.

Contribution to the Project

My PCA visuals and dashboard planning prepared the project for the main dashboard development in the following week.

Challenges Faced

PCA results can be confusing for people who are new to the topic. I handled this by studying different presentation methods and choosing a simple visual style that explains the results without too many technical terms.

Plan for Next Week

In Week 8, I will build the interactive dashboard using Streamlit and Plotly and bring all the charts together.

Weekly Summary

Week 7 prepared the PCA visuals and the dashboard plan. This made the dashboard construction in the next week much easier.